

Presented by :
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Customer Satisfaction Analysis



Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Project Objective

Objective

The objective of this project is to analyze customer support interactions and identify factors affecting customer satisfaction (CSAT). A Machine Learning model is developed to predict customer satisfaction scores based on customer support and operational data.

Goals

- Perform Exploratory Data Analysis (EDA)
- Identify important factors affecting CSAT
- Build a predictive Machine Learning model
- Evaluate model performance



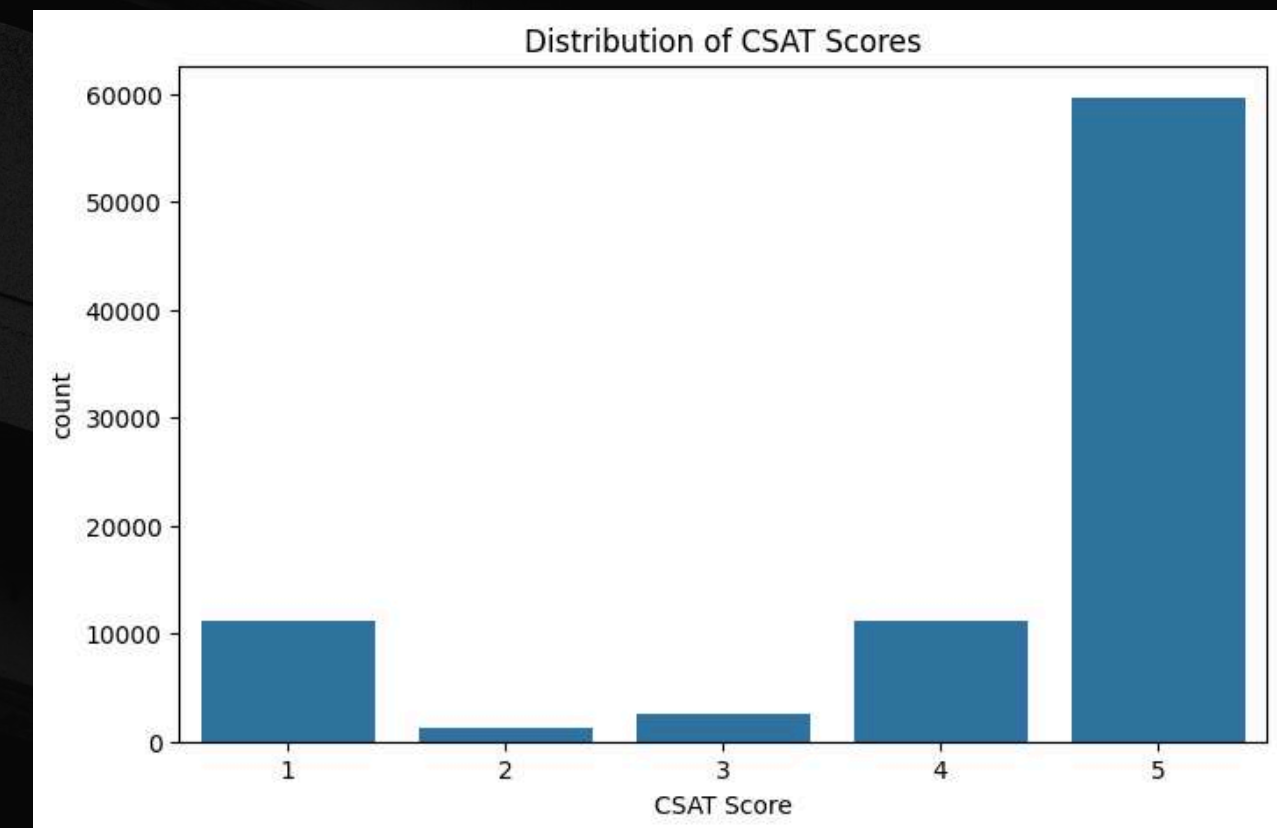
Dataset Overview

Sample Features:

- Channel Name
- Category
- Sub-Category
- Agent Name
- Supervisor
- Manager
- Agent Shift
- Item Price
- Issue Reported Time
- Issue Response Time

Dataset Information

- Total Records: 85,907
- Original Features: 20
- Target Variable: CSAT Score



CSAT Scores:

Customer satisfaction ratings from 1 to 5.

Data Preprocessing



Data Cleaning & Preprocessing



- Removed irrelevant columns:
 - Unique ID
 - Order ID
 - Connected Handling Time

- Removed columns with excessive missing values:
 - Customer Remarks
 - Customer City
 - Product Category
 - Order Date Time
- Filled missing Item Price values using Median.
- Converted date columns into datetime format.
- Encoded categorical variables using Label Encoding.

Exploratory Data Analysis

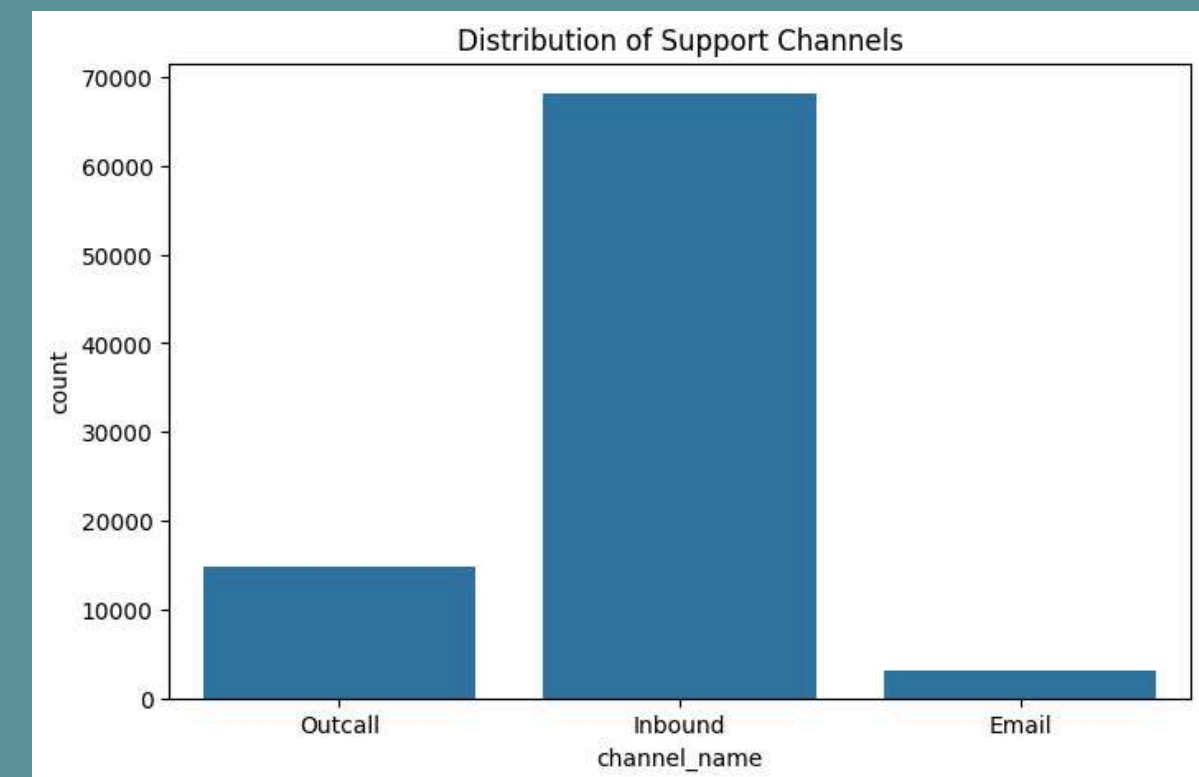
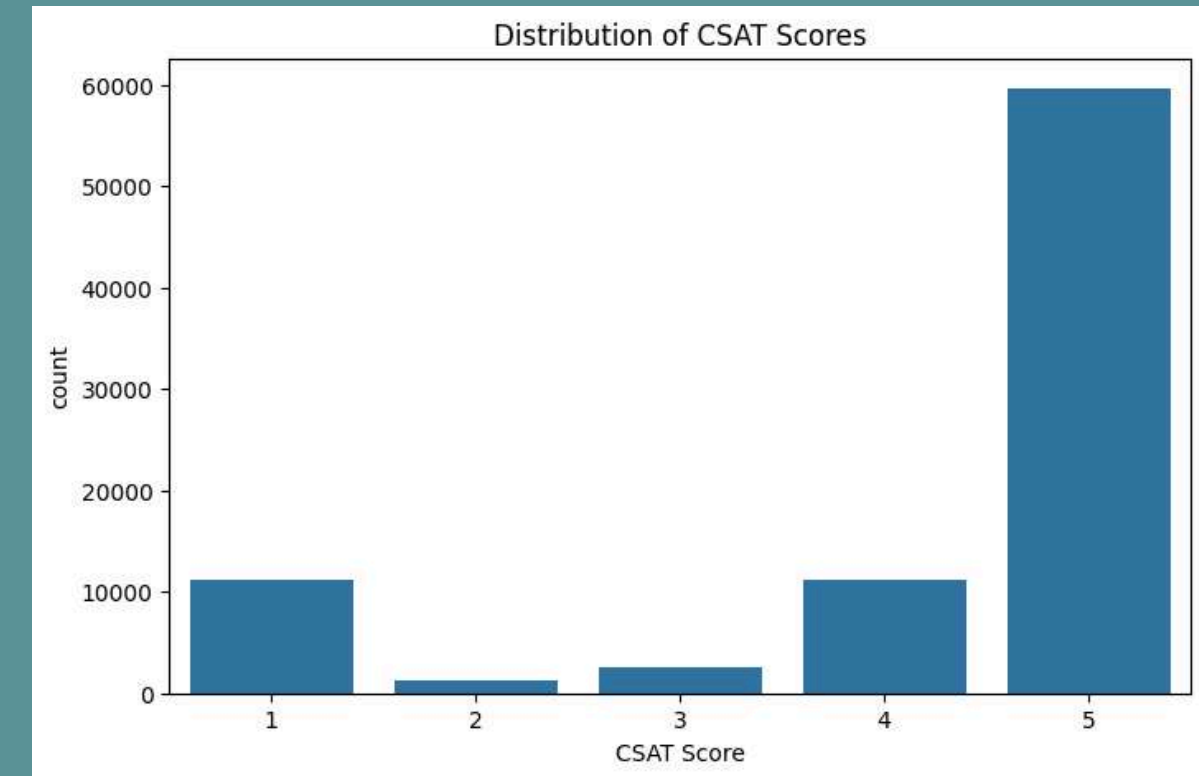
Key Insights from EDA

Customer Satisfaction Distribution

- Majority of customers provided a CSAT score of 5.
- Dataset is highly imbalanced.

Channel Analysis

- Inbound channel handled most interactions.
- Outcall and Email contributed fewer interactions.

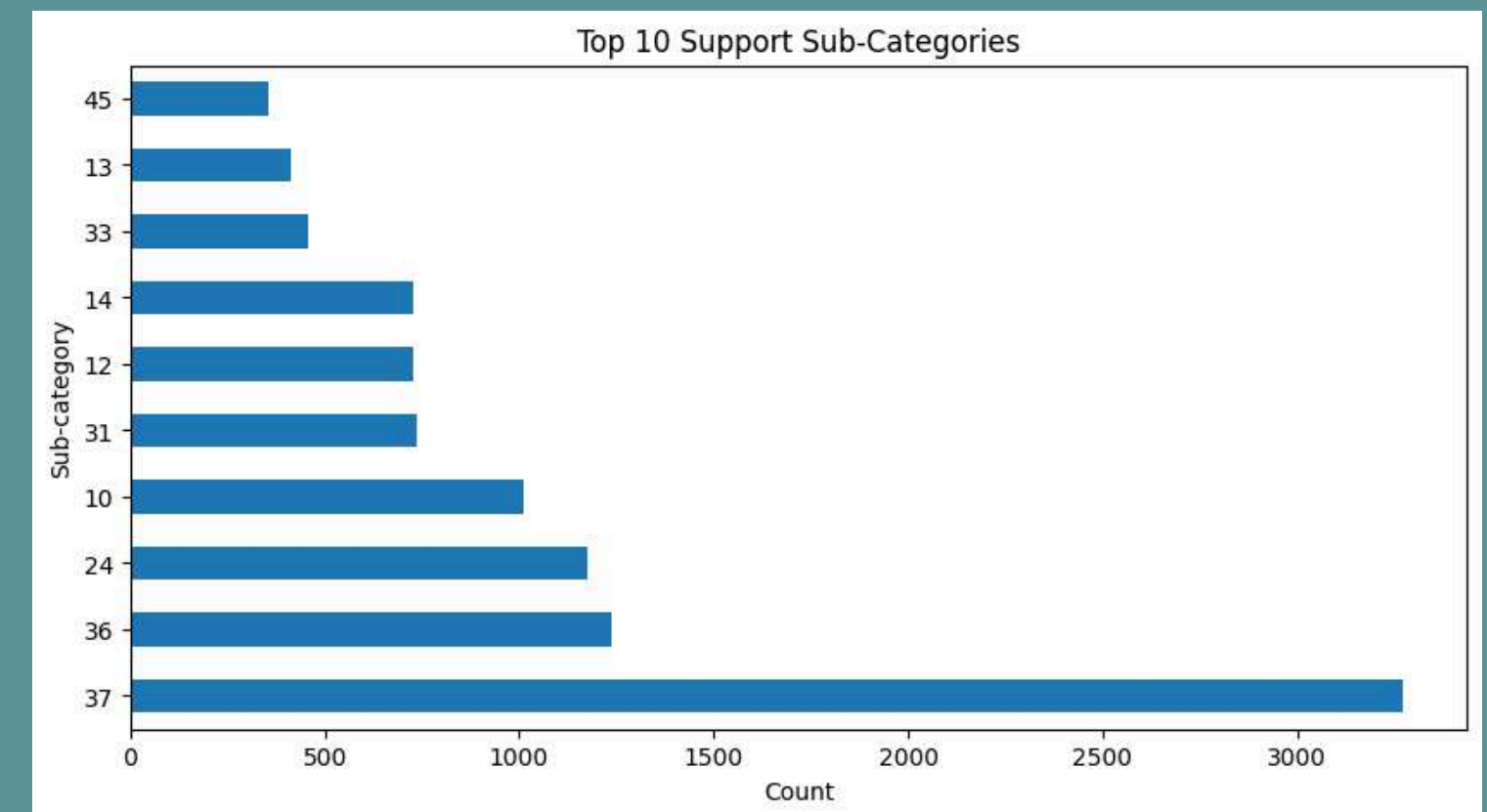
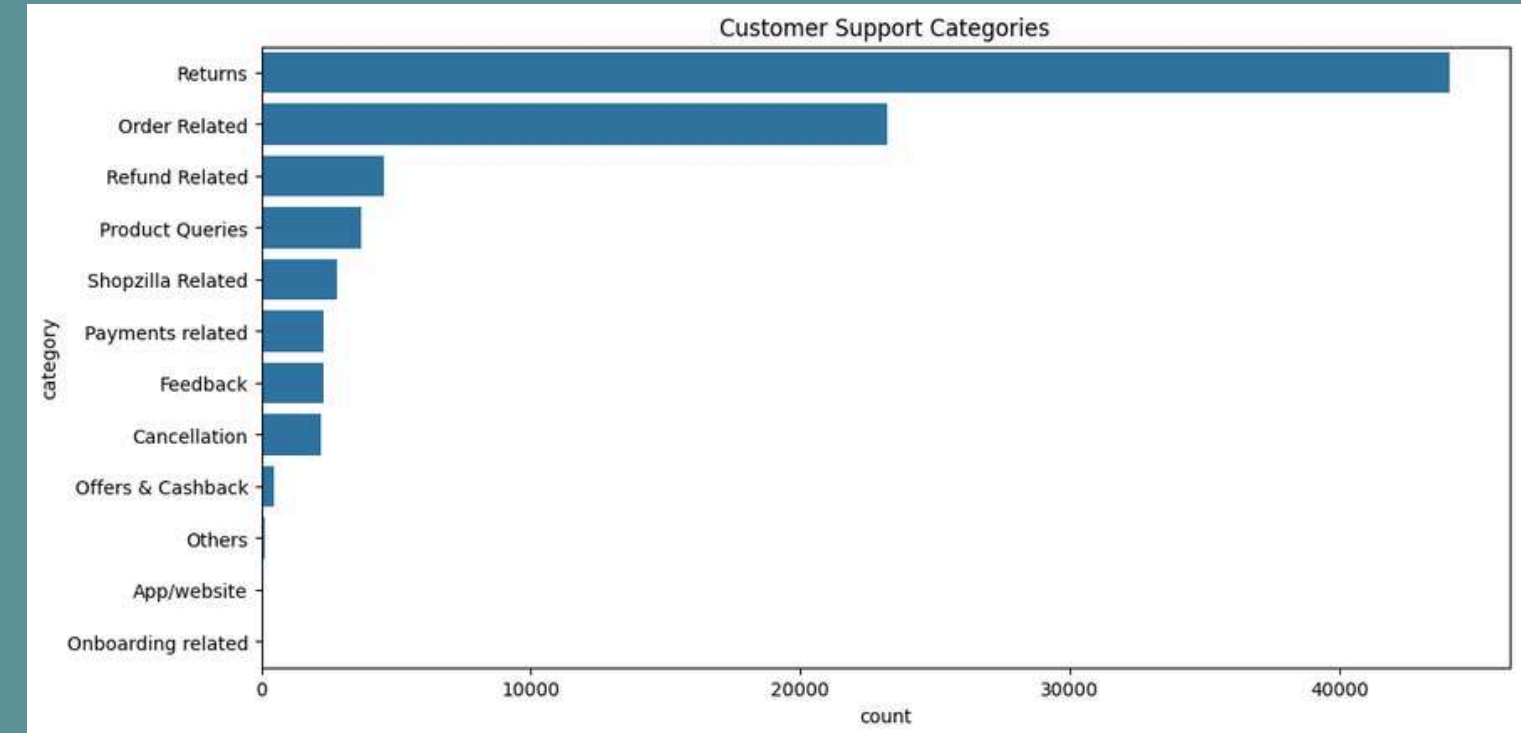


Exploratory Data Analysis

Category Analysis

Most common categories:

- Returns
- Order Related
- Refund Related



Feature Engineering

A new feature was created:

Resolution_Time_Hours

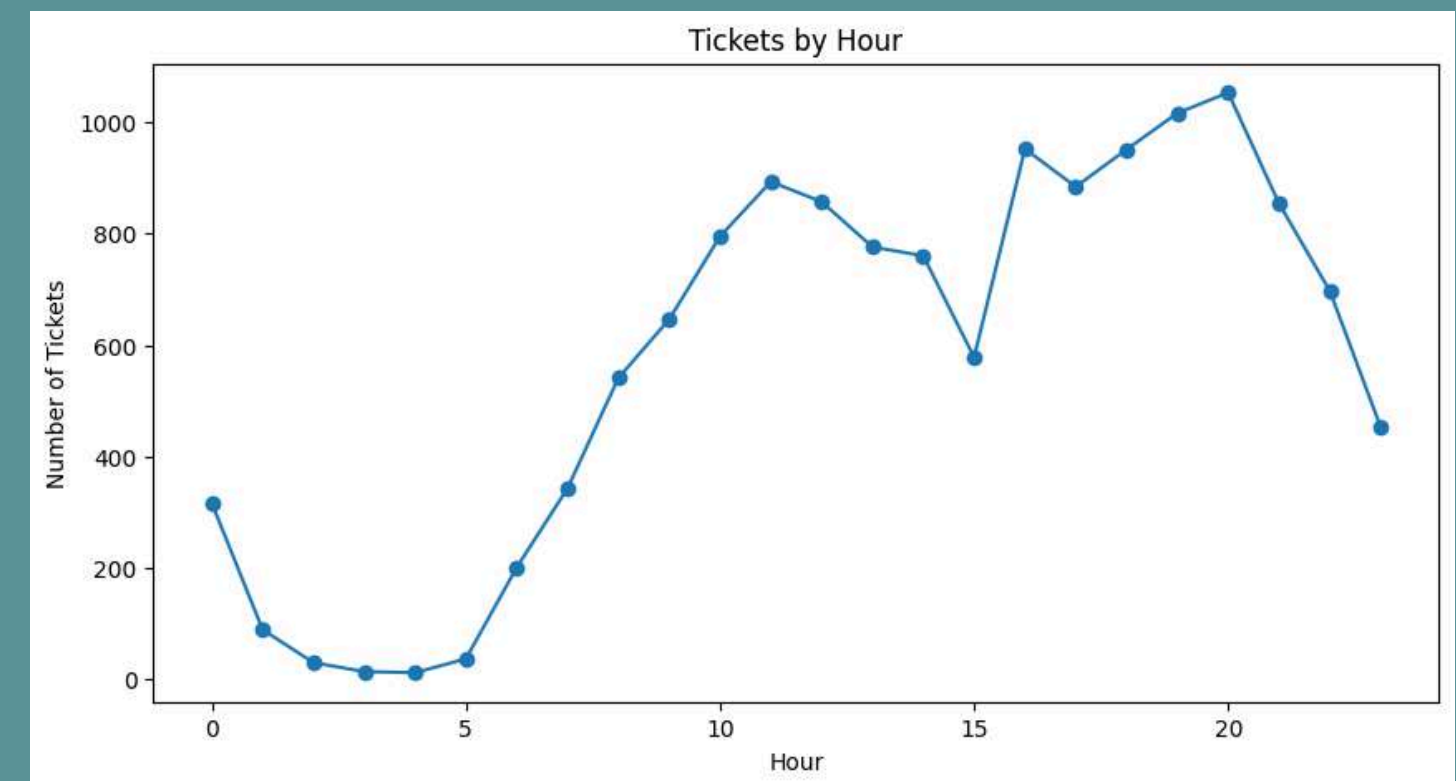
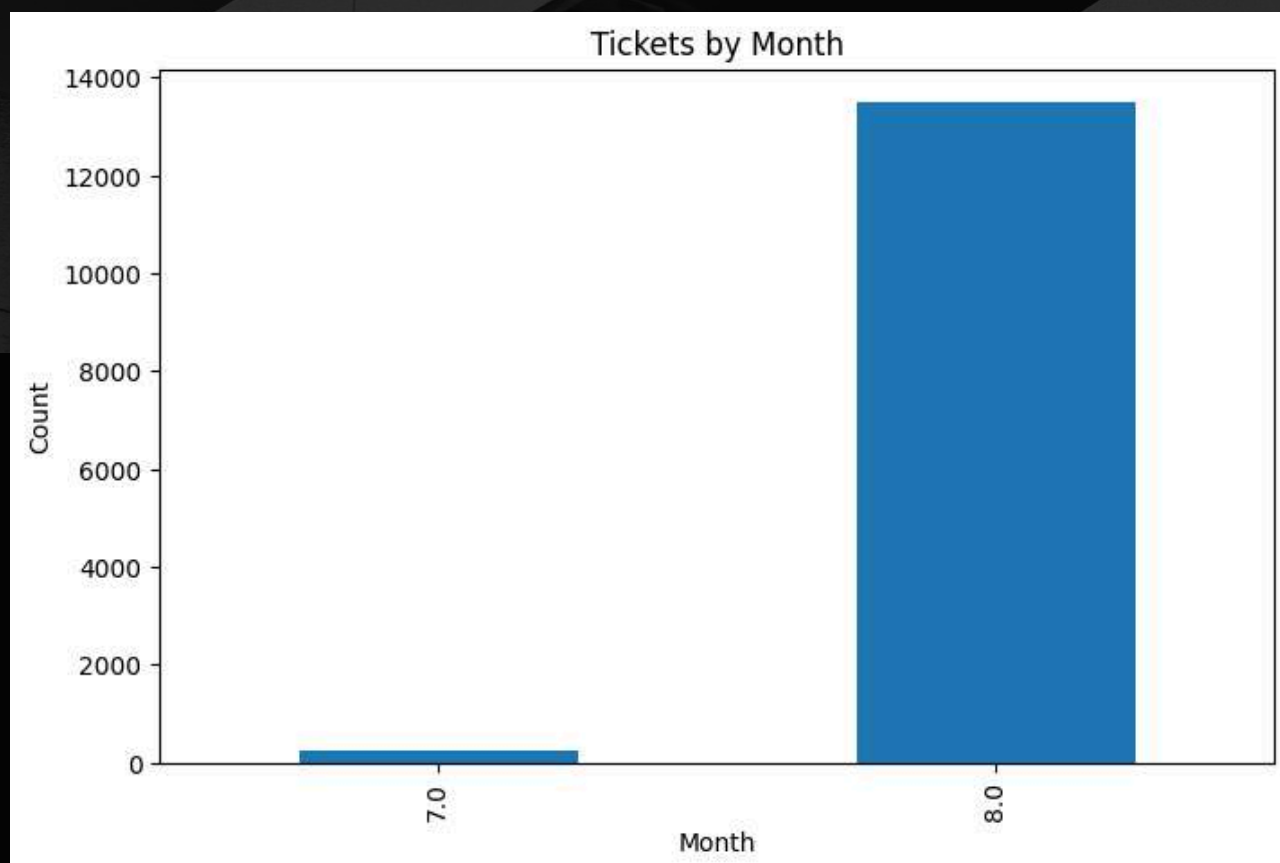
Resolution Time = Issue Resolved Time – Issue Reported Time

Additional Features Created

- *Report_Hour*
- *Report_Day*
- *Report_Month*

These features help analyze customer behavior based on time.

This feature measures how quickly customer issues were resolved.



Feature Engineering

A new feature was created:

Resolution_Time_Hours

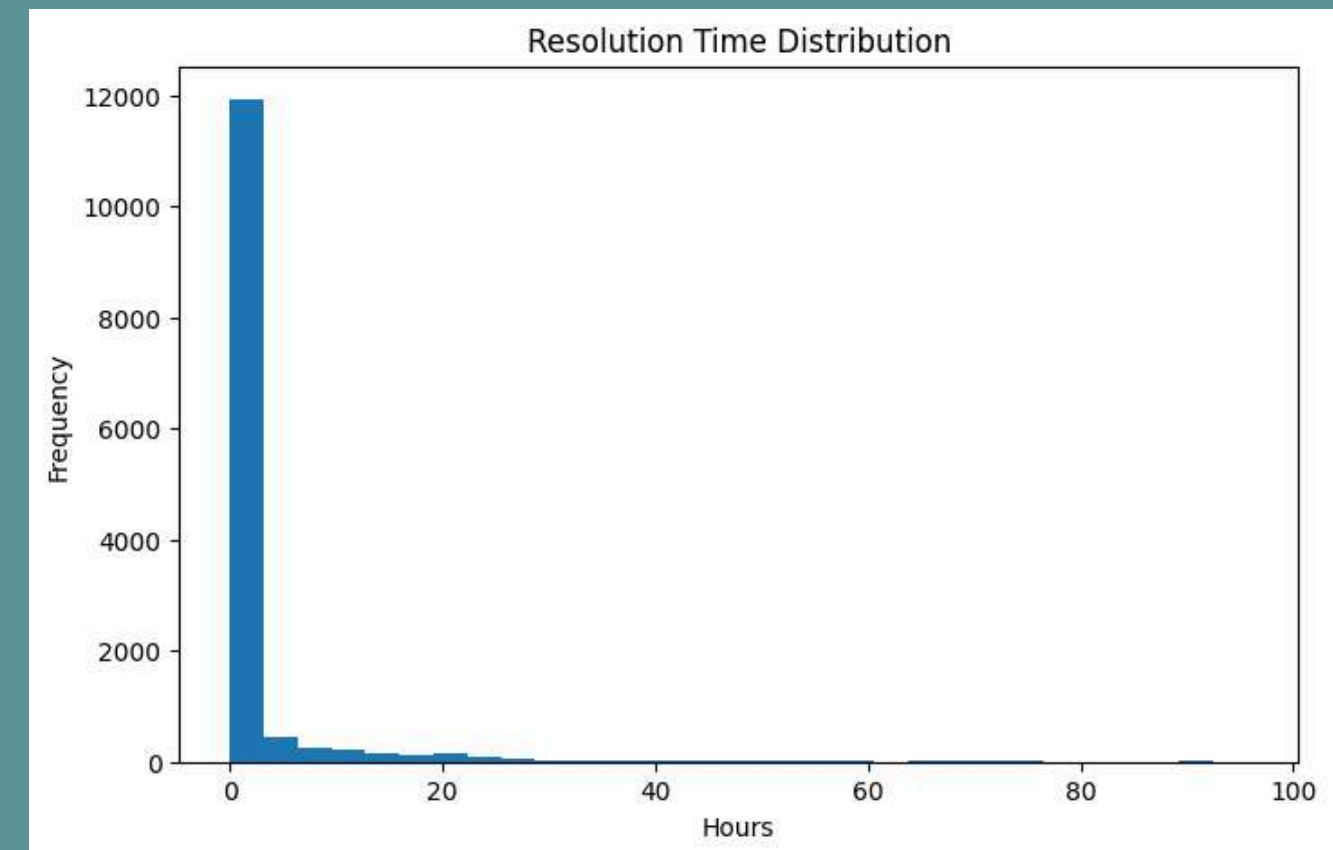
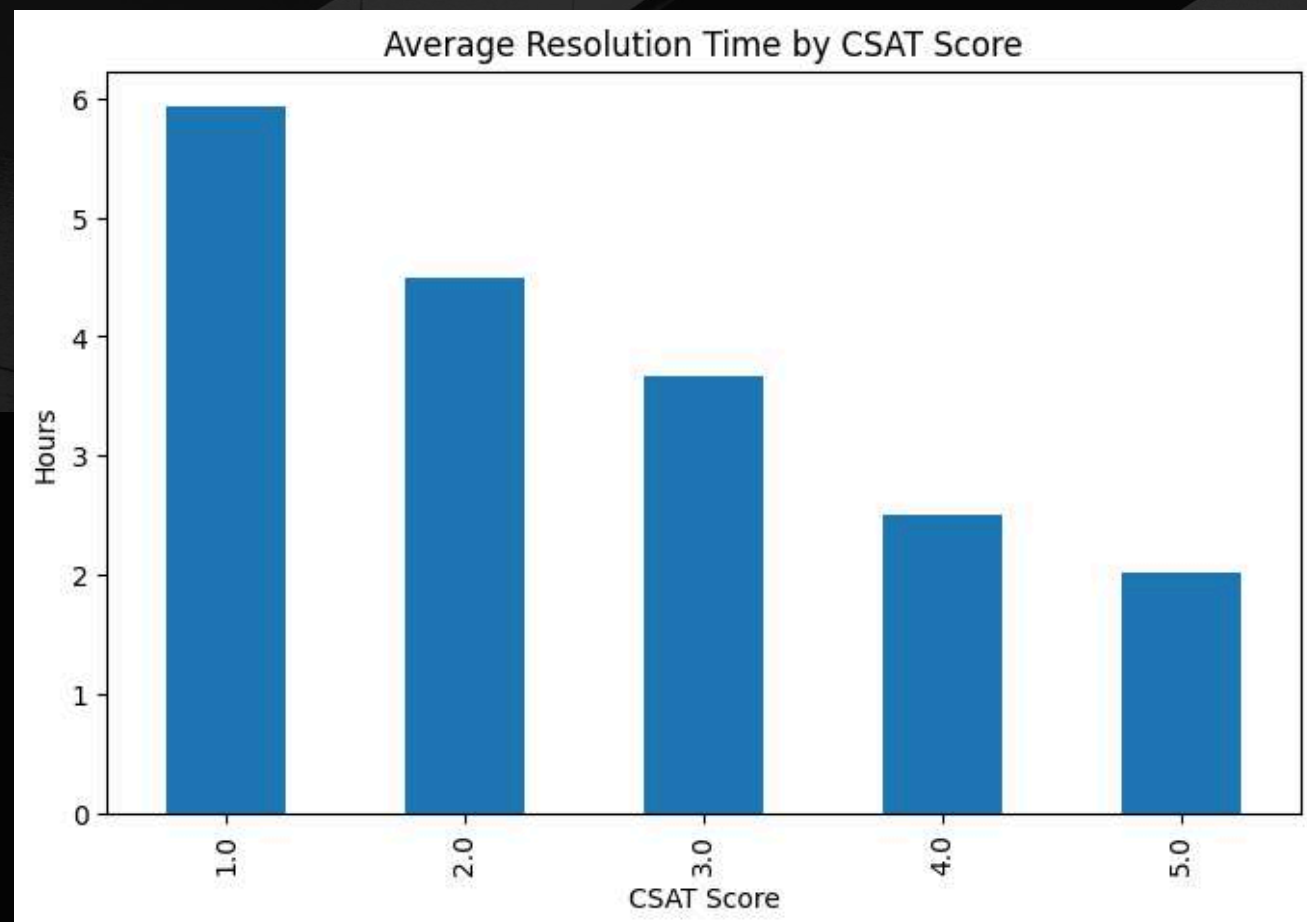
Resolution Time = Issue Resolved Time – Issue Reported Time

Additional Features Created

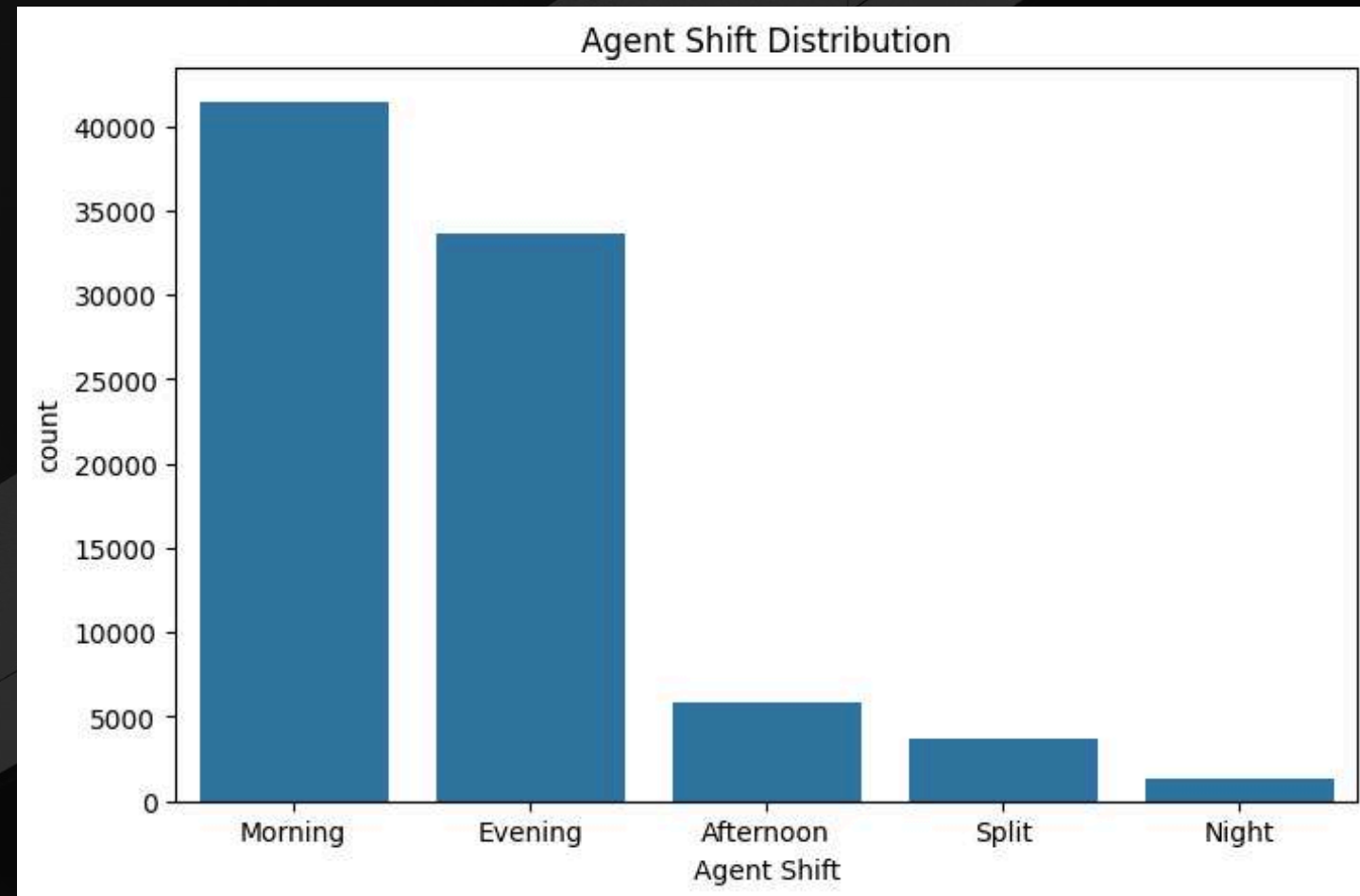
- *Report_Hour*
- *Report_Day*
- *Report_Month*

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Machine Learning Model



Algorithm Used:
Random Forest Classifier

Why Random Forest?

- Handles categorical and numerical data effectively
- Reduces overfitting through multiple decision trees
- Provides feature importance analysis

DATA SPLIT:

Training data: 0%
Testing data: 20%

Model Performance

Evaluation Results

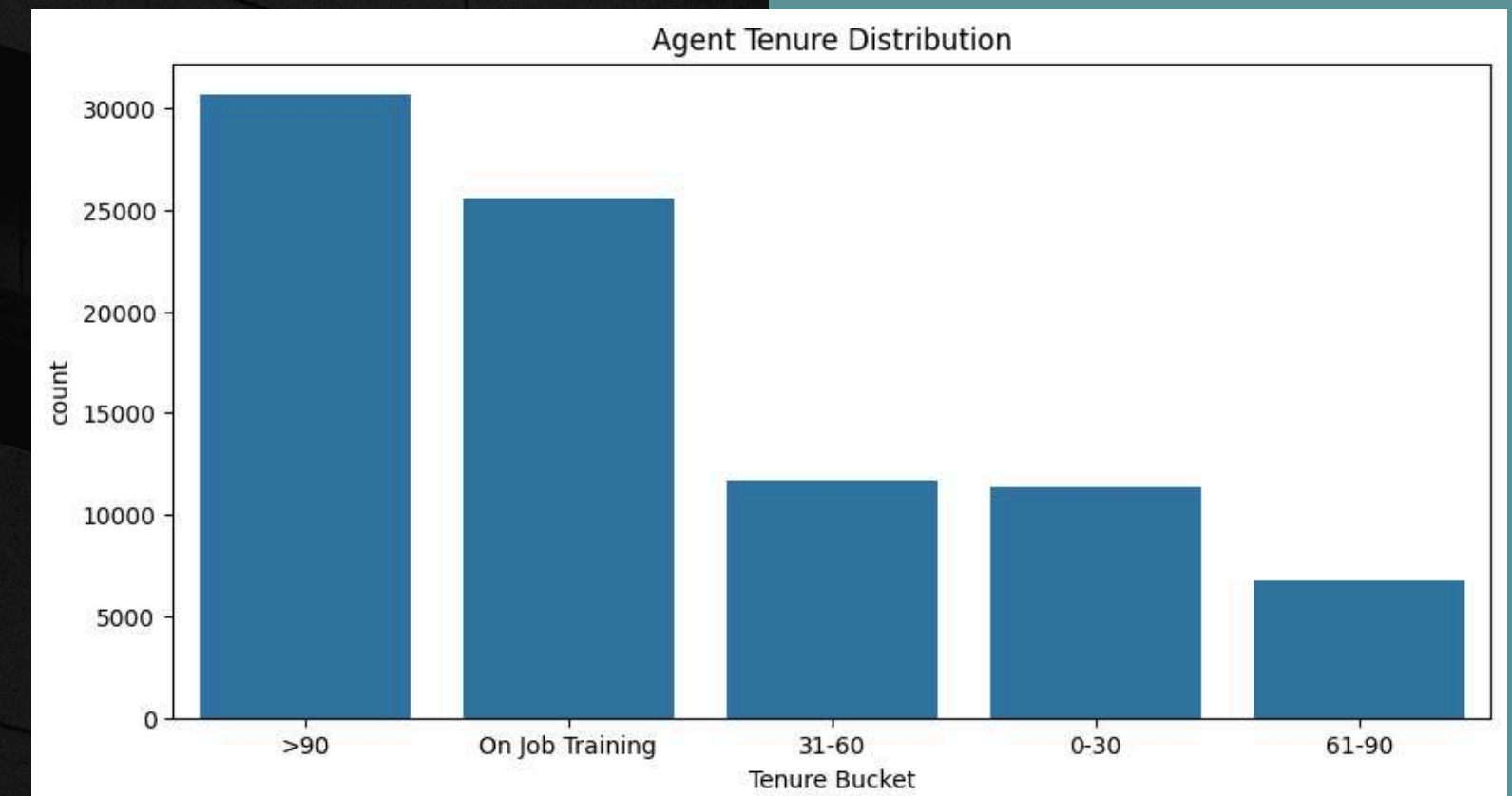
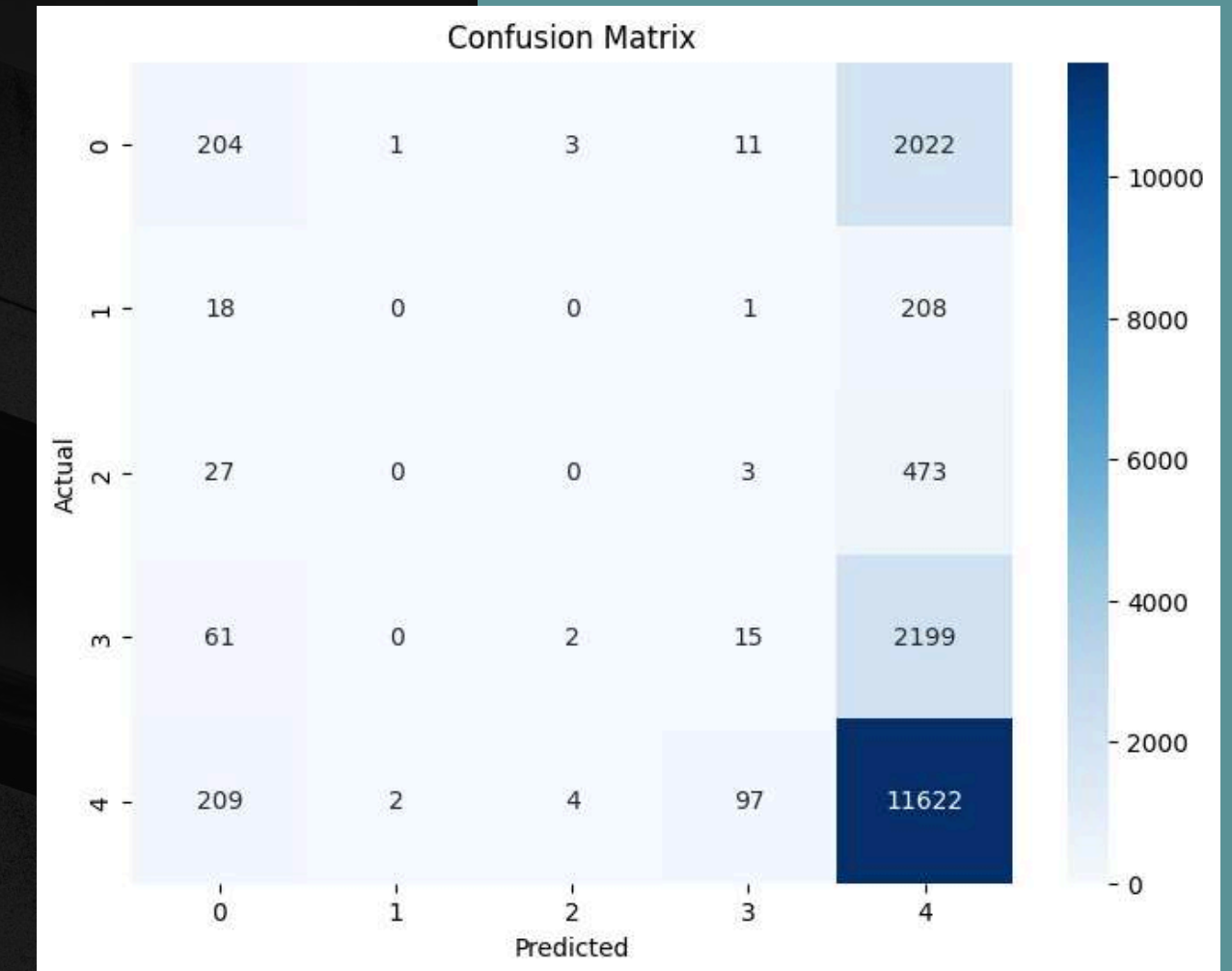
Accuracy - 68.9%

Classification Metrics

The model performs best for CSAT score 5 due to class imbalance in the dataset.

Observation

Most customer records belong to CSAT score 5, making prediction of lower ratings more challenging.



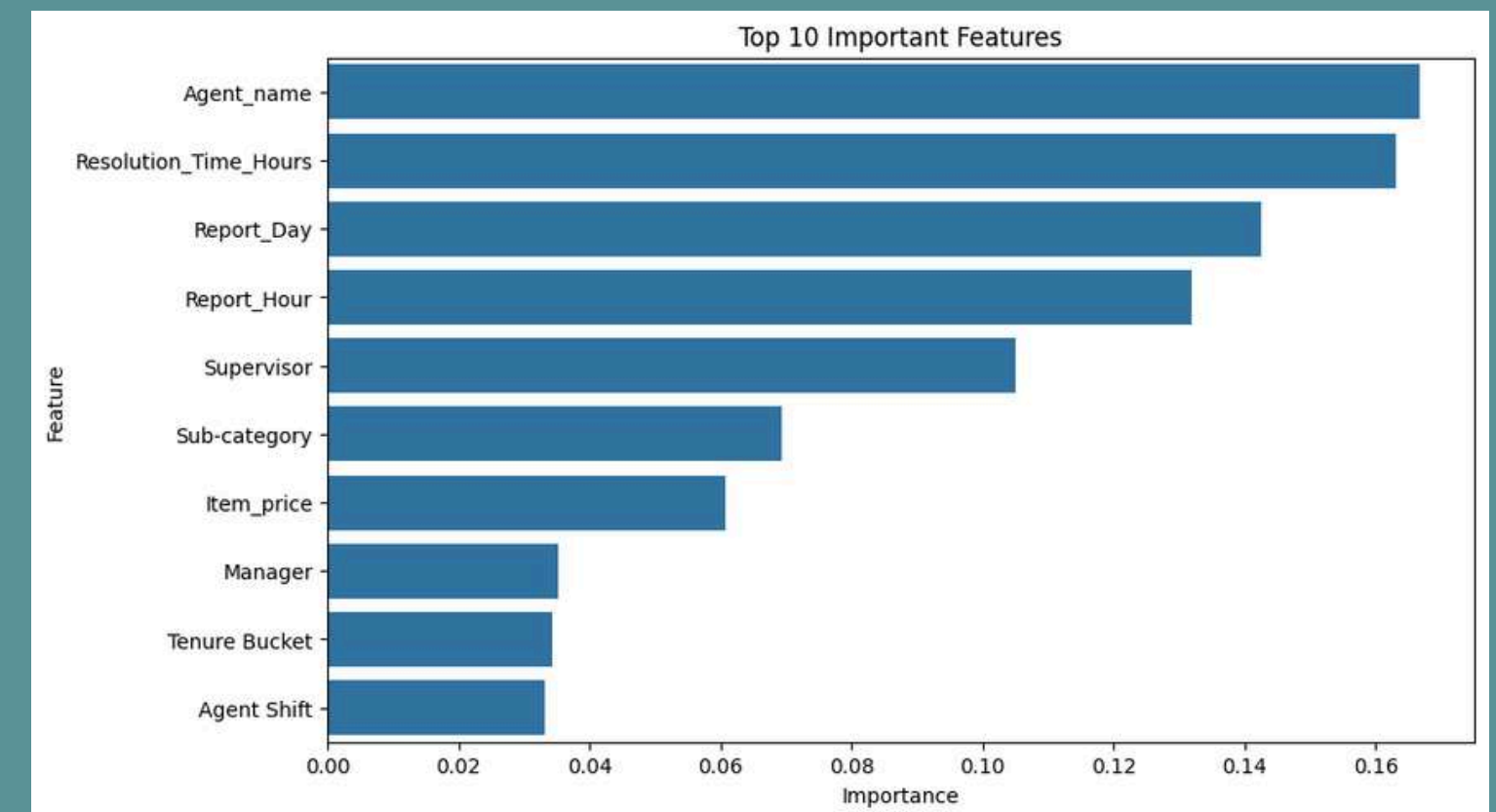
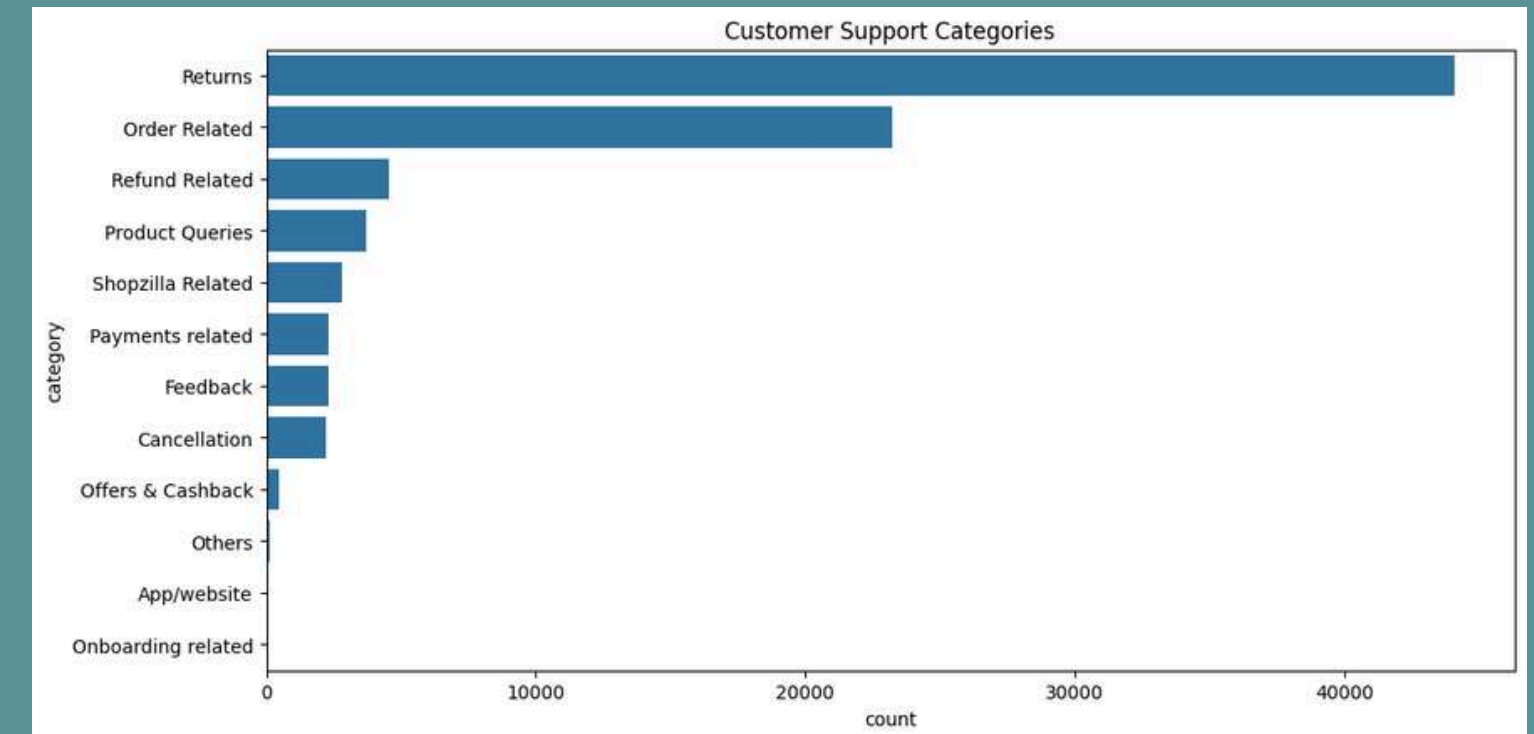
Feature Importance

According to the Random Forest

1. *Agent Name*
2. *Resolution_Time_Hours*
3. *Report_Day*
4. *Report_Hour*
5. *Supervisor*
6. *Sub-Category*
7. *Item Price*

Interpretation

- *Faster resolution times improve customer satisfaction.*
- *Agent performance influences customer experience.*
- *Time-related factors also affect satisfaction levels.*



Conclusion

This project successfully analyzed customer support interactions and customer satisfaction scores.

Final Result

Random Forest achieved an accuracy of 68.9%.

Key Findings

- Most customers reported high satisfaction.
- Returns and Order Related issues were the most common complaints.
- Resolution time significantly impacts satisfaction.
- Agent-related factors strongly influence CSAT scores.





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Thank You
